Naming the Magic · IG · Practice

U05a · 24 written + 6 pen-first = 30 marks · Name: _____

Fresh authored questions. Work on paper first. Use pseudocode only where requested; code labelled Python uses Python. Assume valid input where stated. No computer is required to complete the sheet.

1. Ours: choose a type

Give a suitable basic data type for each item:
(1) Number of visitors: 18
(2) Donation in currency units: 4.75
(3) Whether the door is open: TRUE
(4) One-letter shelf code: 'R' **[4]**

2. Ours: declare, then assign

Write four pseudocode statements: declare Banner as STRING and assign "Blue lantern"; declare Symbol as CHAR and assign 'L'. **[2]**

3. Ours: what may change?

Distinguish a variable from a constant in terms of their values during program execution. **[2]**

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4. Ours: read the changing state

```
score = 3
bonus = score + 4
score = bonus + 4
bonus = score + 4
print(score, bonus)
```

(a) Trace lines 2, 3 and 4. After each line, write its line number and the values of both score and bonus. Repeat unchanged values. **[3]**

(b) Write the exact output of line 5. **[1]**

5. Ours: the input trap

```
count = input("Count? ")
print(count + 2)
```

(a) The user enters 6. State the Python type of count. **[1]**

(b) Explain why the second line fails. **[1]**

(c) Rewrite only the first line so the addition works for valid whole-number input. State the output after entering 6. **[2]**

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6. Ours: make the next edit easier

A bicycle-hire program repeats the hourly rate, 9, in several calculations. A developer proposes the meaningful name HourlyRate.

(a) Write the IG pseudocode declaration that defines HourlyRate as a constant with value 9. **[1]**

(b) Explain two benefits of replacing the repeated literal 9 with that named constant in this program. **[2]**

(c) State one way a useful comment helps someone maintain this program. **[1]**

7. Ours: the model and the language

(a) Distinguish CHAR from STRING in exam notation. **[2]**

```
LIMIT = 12      # INTEGER
LIMIT = "open"
print(LIMIT)
```

(b) State the exact output. Explain why neither the uppercase name nor the type comment prevents this result in Python. **[2]**

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8. Ours: the badge desk · pen first

💻 **6 marks · design before code.** A badge desk charges 7 currency units per badge. Input a visitor label (text), then a badge count (a valid non-negative whole number). Store the fixed unit price using a named constant convention, calculate the total and output the visitor label and total. No decisions or loops are needed.
Optional after writing: type Q8(b) into `challenge_U05a_TheBadgeDesk.py` and run it. All six marks are human marked; a computer is not required.

(a) Write numbered steps or draw a flowchart for this sequence, from inputs through calculation to output. **[2]**

(b) Write the Python program. Use meaningful identifiers and clear output. **[3]**

(c) For visitor label Guest and badge count 4, state the label and total your program must output. **[1]**

Above and beyond · outside the 30 marks. Start with `a = 5` and `b = 13`. Using assignments and a third name, swap the values without losing either original value. Trace your steps. Why does `a = b; b = a` fail?

Additional space - label the question number.